Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/18/2018
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Project Title:	Enhancement and Verification of Quantitative Phase-Field Crystal Modeling using NASA-PSI Coarsening in Solid-Liquid Mixtures Experiments Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Laradji, Mohamed Ph.D. (University of Memphis)		
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Task Description:	Verification and enhancement of predictive computational material models for solid-liquid mixtures based on experiments conducted on Earth is challenging because of the detrimental gravitational effects on solid particles and liquid mixtures, e.g., buoyancy-induced convection and particle sedimentation. Microgravity environment for NASA-PSI (Physical Sciences Informatics) “coarsening in solid-liquid mixtures experiments” provides data without the detrimental gravitational effects that can be used for verification and enhancement of predictive models. In return, the verified/enhanced model can be used to predict coarsening in solid-liquid mixtures that ideally eliminates the need to conduct similar experiments for other materials and also experiments for other phenomena involved with solid-liquid coexisting. The mentioned predictive capability is only possible if the computational model is not derived phenomenologically for the specific application and material; phase-field crystals (PFC) are a class of such predictive computational models. Therefore, the objectives of this proposal are: 1) The development of a quantitative multi-time and length-scale computational modeling framework and tool around PFC to predict coarsening in Pb-Sn solid-liquid mixtures. Challenges for this objective are quantifying the PFC model and developing an efficient computational tool for three-dimensional (3D) simulations. 2) Data mining of NASA-PSI Coarsening in Solid-Liquid Mixtures (CSLM), CSLM-2, CSLM-2R, and CSLM-3 experiments and using them in series of simulations in the developed computational tool for enhancement and verification of the proposed modeling framework. The challenge for this objective is the huge number of 3D simulations that require significant amount of computer power. 3) Quantifying the developed model and tool to Al-Cu solid-liquid mixtures to show the potential benefits of the proposed research. The challenge for this objective is only the quantification process of the model.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed computational tool verified/quantified based on CSLM data will establish a virtual laboratory that can be used to study coarsening as well as other material processing phenomena on Earth involving with solid-liquid coexistence; e.g., rapid solidification and grain growth. Since computational models are highly repeatable at lower cost, the proposed modeling framework will result in accelerated material process development and certification. Since solid-liquid coexistence is the initial step in many manufacturing technologies such as casting, liquid infiltration, and additive manufacturing, this modeling framework can be used to investigate the relations between these manufacturing processes and the material properties in order to optimize/control the manufacturing processes parameters. In addition, the results of this research will be disseminated to the public in forms of peer-reviewed articles, scientific reports, a project-specific website, and data uploaded to NASA-PSI website.
Task Progress:	Our central goal is to use NASA-PSI CSLM experimental data for verification and enhancement of a multi time and length scale computational modeling framework and tool. The proposed multi-scale modeling framework is established around a novel and predictive computational material model (phase-field crystals, PFC), which is a unique computational model with the capability of simulating solid-liquid processes on diffusive time scales with atomistic resolutions. Key objectives are to 1) Develop and quantify binary PFC models for coarsening of solid-liquid mixtures; 2) Develop a highly-parallel semi-implicit spectral and operator splitting numerical algorithm to use on high performance computing facilities that will enable the necessary large-scale PFC simulations (comparable to CSLM data length-scale); 3) Use molecular dynamics (MD) simulations to calculate input material properties required for quantification of PFC model; and 4) Ensure accuracy and reliability of MD calculations by using and/or developing one of the most reliable and complicated MD interatomic potentials in our simulations; i.e., modified embedded-atom method (MEAM). The proposed multi-scale modeling is calibrated for Pb-Sn system which is the material used in CSLM experiments. Once the multi-scale modeling framework is validated, we will calibrate the modeling framework for other binary systems to show versatility of the proposed multi-scale modeling framework to simulate the coarsening phenomenon for other materials. The research team has consistently met bi-weekly since January 1, 2018 to review and discuss progress on project tasks. Accomplishments for year 2018 to date are as follows: 1) We developed a binary Pb-Sn interatomic potential based on the MEAM formulation. Our MEAM potential for Pb improves the MP calculation by 17%, without any considerable compromise in other calculated properties. Furthermore, our results indicated that the reparametrized MEAM potential for Pb also predicts improved thermodynamic properties such as the latent heat and expansion in melting and higher accuracy in the values of the elastic constants at high temperatures. The scenario for Sn is more complex. In this case, we needed not only to achieve an accurate MP, but also a stable ß-Sn structure near the MP. We noted that the Ravelo-Baskes MEAM potential [1] indeed predicts a stable ß-Sn phase, but it predicts that the melting point (MP) is around 31% lower than the experimental value. We also showed that the previously developed MEAM potential for Sn, by Vella et al. [2] cannot be used for simulations of coexisting solid and liquid phases as well as for CSLM, because it predicts an unstable ß-Sn phase at high temperatures. We therefore, considered the MEAM parameters of Sn, rendered earlier by Ravelo and Baskes [1], and adjusted its parameters such that an accurate MP is predicted along with a stable ß-Sn at high temperatures. The newly parameterized MEAM potential for Sn results in elastic constants at 0 K and high temperatures that are in good agreement with experiments. However, we showed that the elastic constants in the solid phase are sacrificed at the expense of better accuracy of the model around the MP. Based on the new MEAM parameterization of elemental Pb and Sn, we developed MEAM parameters for binary Pb-Sn alloys fitted to the available experimental data, namely the enthalpy of mixing and lattice parameters of Pb-rich solid solution at low temperatures and ß-Sn as well as the enthalpy of mixing of Pb-Sn liquid solutions at high temperatures. Finally, to test the transferability of the present binary MEAM potential for high temperature applications, we used hybrid MD/Monte Carlo (MD/MC) simulations in the semi-grand canonical ensemble for the calculation of the solidus and liquidus lines in the Sn-rich portion of the phase diagram, and showed close agreement with experiments and CALPHAD data. The Sn-rich portion of the phase diagram is identical to those processes at NASA CSLM experiments. Therefore, we have developed a reliable MD model to provide input properties to our PFC model. For instance we calculated liquid structure factor of Pb-Sn mixtures at different concentrations showing that S(k2)/S(k1) and S(k3)/S(k1) ratio was maximum at Pb-0.6Sn. The liquid structure factor is one of the properties that will directly connected to PFC model. The result of this important accomplishment was published [3] at Acta Materialia which is a very prestigious journal in

the field with Impact Factor of 6.036. 2) We investigated the prospect of using PFC model for pure system to study CSLM. Therefore, we investigated the kinetics of Ostwald ripening in two dimensions from a numerical simulation of the single mode phase field crystal model. We showed that this model properly describes the kinetics of Ostwald ripening, with an average domain size, $\bar{R}(t)$ that grows with time as t1/3, in agreement with the classical LSW theory and CSLM data. The domain size distribution, however, is found to be symmetric and is well fitted by a Gaussian, in disagreement with LSW theory which predicts a highly non-symmetric distribution. The domain distributions from the present simulations are, however, in very good agreement with an early experimental study by Seul et al. [4] of the kinetics of Ostwald ripening of a binary phospholipidcholesterol Langmuir film. We also confirmed that the systems reached dynamical scaling during late times, as demonstrated by scaling of the density structure factor. Our investigation, of the topological structure of the two-dimensional system during Ostwald ripening, inferred from the Voronoi analysis, indicates that the domains are positioned in space so as to maximize the configurational entropy of the domains centers of mass. In particular, we showed that the average size of the q-coordinated domains follows Lewis's law, i.e., domains with high coordination number are larger than the average domain size, and vice versa. Furthermore the average coordination number of neighboring domains of a given q-coordinate number follows Aboav-Weaire law, i.e., domains with high coordination are surrounded, on average, by domains with low coordination number. In other words, the system adopts a structure such as deviation from a topological charge Q = 0 is minimized. Our results on the topological structure of the coarsening sys-tem are in very good agreements with earlier experimental results of Seul et al.

The result of this important accomplishment is currently under review at Physical Review E which is also a very prestigious journal in the field.

3) We have created a webpage (https://) to summarize our accomplishments in this project and raise the public awareness to NASA-sponsored research. In addition, we have created tutorials for the simulations that we have performed and posted them at a specific webpage (https://) to facilitate further research in this area by the scientific community.

References

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Abstracts for Journals and Proceedings	Nourian-Avval A, Asadi E. "A Quantitative Phase-Field Crystal Model to Study Particle Coarsening in Binary Systems." Symposium: Frontiers in Solidification Science and Engineering. TMS 2018. 147th Annual Meeting, The Minerals, Metals and Materials Society, Phoenix, AZ, March 11-15, 2018. TMS 2018. 147th Annual Meeting, The Minerals, Metals and Materials Society, Phoenix, AZ, March 11-15, 2018. http://www.programmaster.org/PM/PM.nsf/ApprovedAbstracts/5C827BD50A4F91868525815D007438FD?OpenDocumen ; accessed 10/30/18. , Mar-2018
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